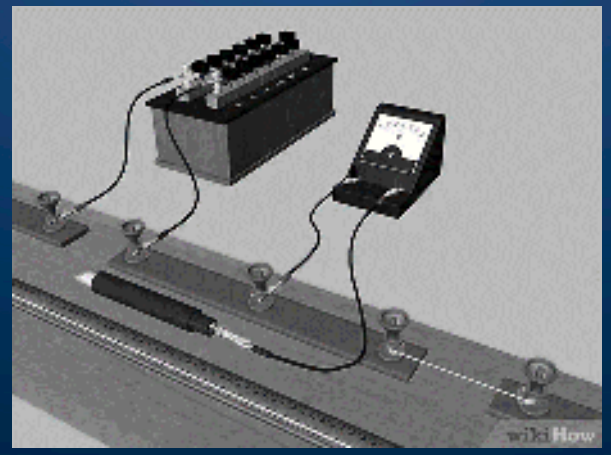
TO DETERMINE THE SPECIFIC RESISTANCE OF THE MATERIAL OF A WIRE BY A METER BRIDGE

EXPERIMENT NO - 7



SUBMITTED TO: DR. REZAUL K. SIDDIQUE

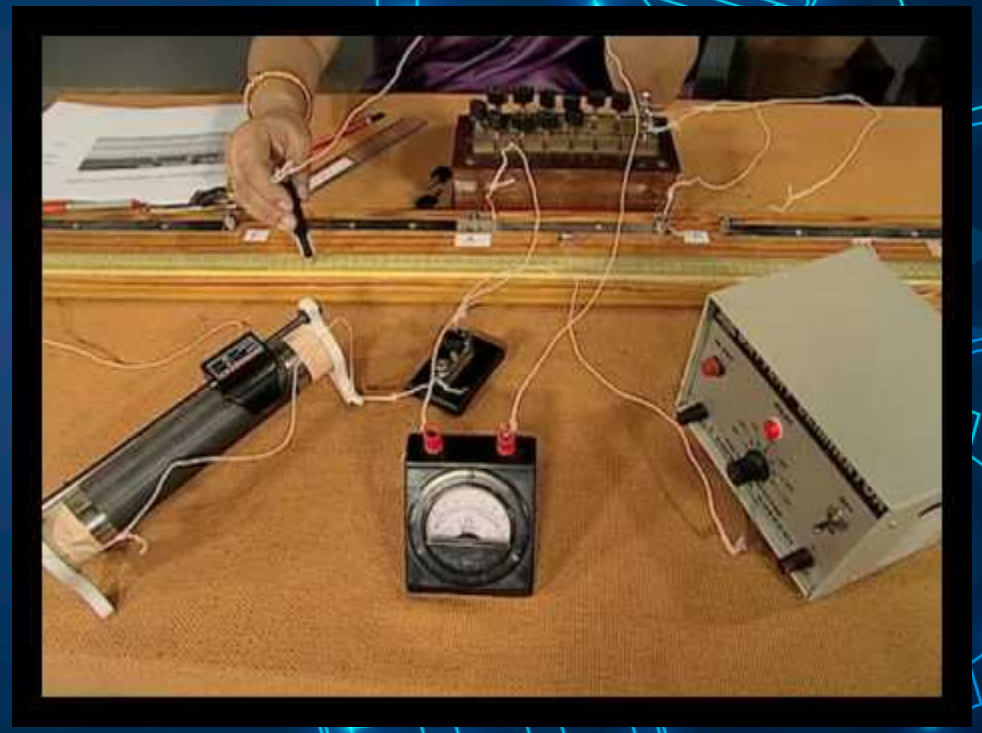


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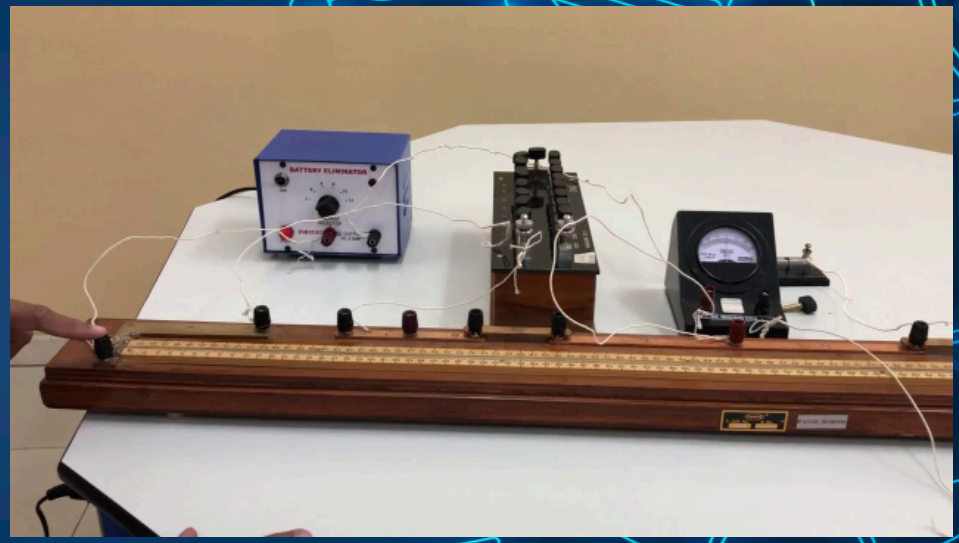
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OBJECTIVE

To find the resistance of a given wire using a meter bridge and hence determine the resistivity of it’s material



THEORY

In the given arrangement, using Wheatstone's principle:

$$\frac{P}{Q} = \frac{X}{R} \quad \text{(Wheatstone bridge circuit)}$$

$$\frac{X}{R} = \frac{l}{100 - L} \quad \text{(Meter bridge circuit)}$$



$$\text{Or, } X = \frac{RL}{100 - L} \qquad (1)$$

When the resistances X and R are interchanged, we get

$$\frac{R}{x} = \frac{RL}{100 - L} \qquad (2)$$



The mean value of (1) and (2) will give the value of the unknown Resistance X.

$$X = \frac{\rho L}{\pi r^2}$$

$$\text{Or, } \rho = \frac{X \pi r^2}{L}$$

Where ρ is the specific resistance of the of the wire. ρ is determined after measuring X , r and L.

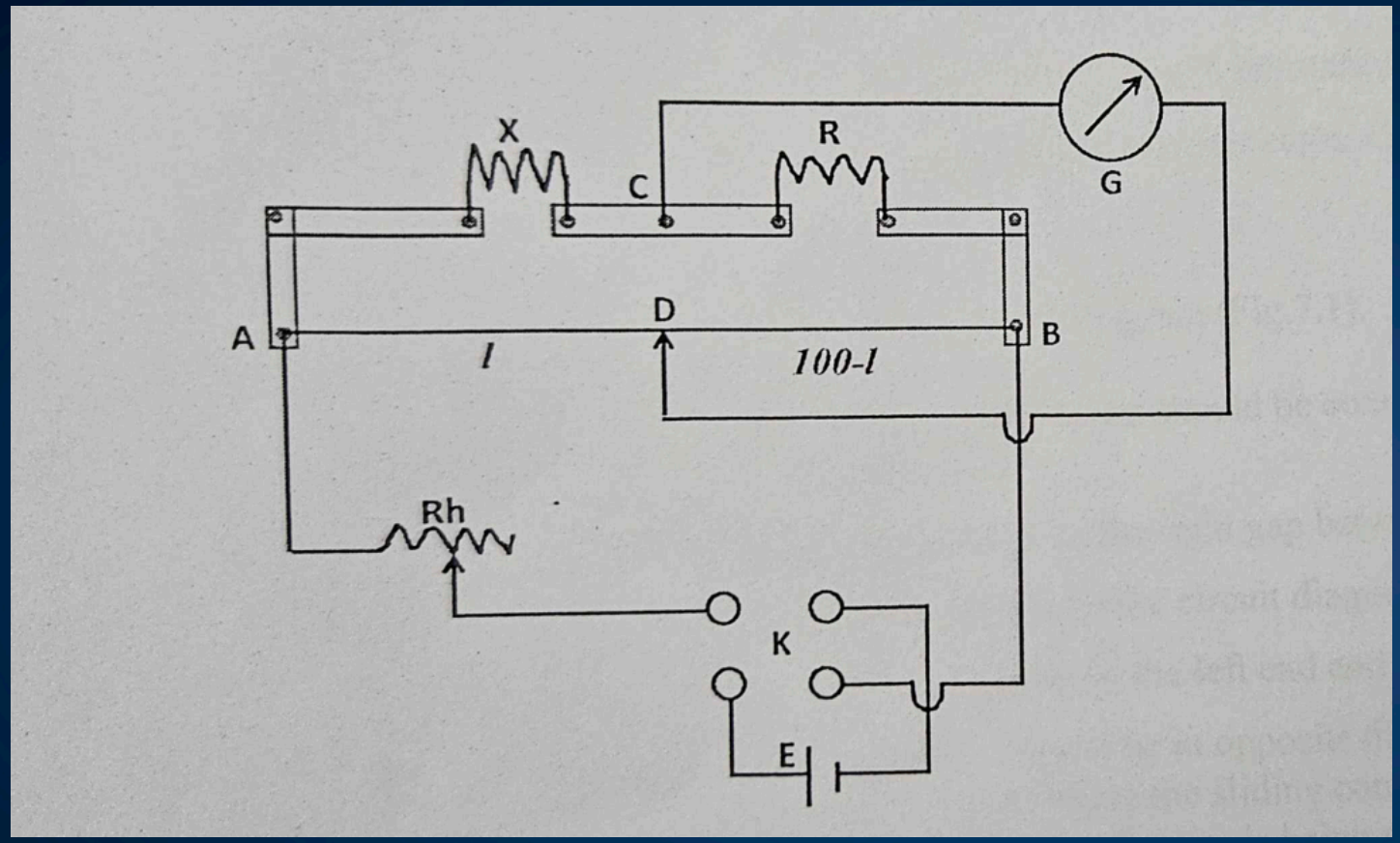


Figure – Circuit diagram for the experimental setup

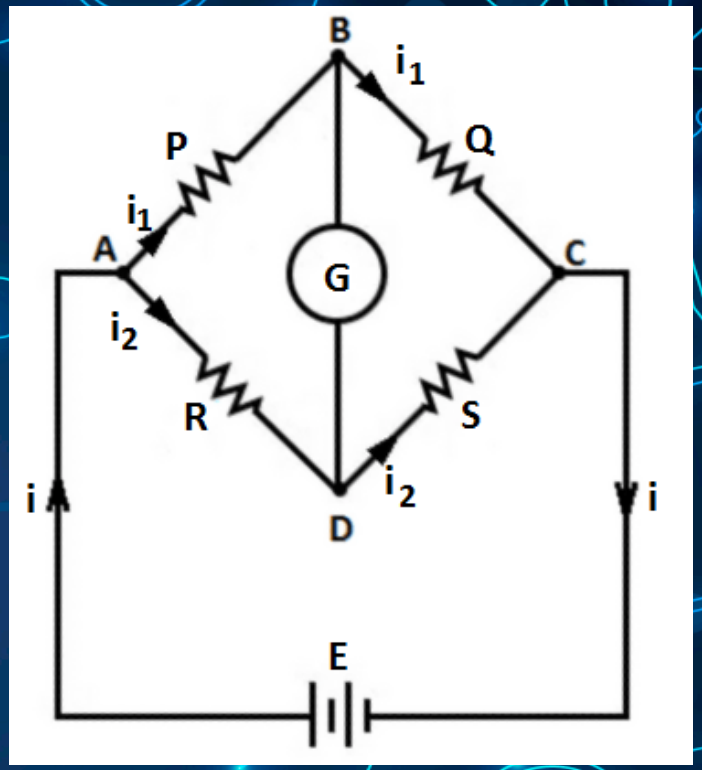


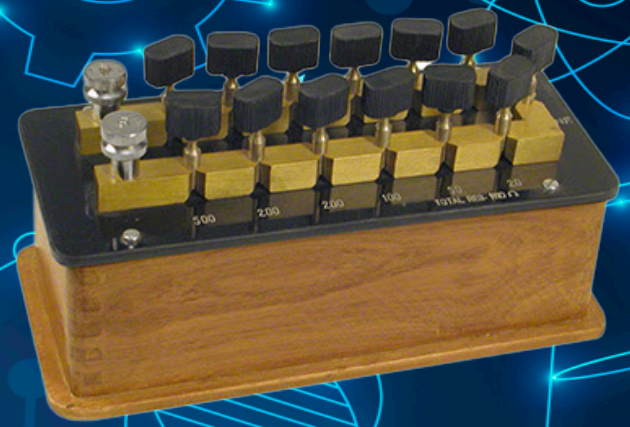
Figure – Wheatstone Bridge

APPARATUS





- METER BRIDGE
- GALVANOMETER
- RESISTANCE BOX
- JOCKEY
- COMMUTATOR
- SPECIMEN WIRE
- SCREW GAUGE
- METER SCALE
- CONNECTING WIRES





PROCEDURE

- The connection should be made according to the circuit diagram.
- The wire is connected in the left gap (A to C) without loops.
- The resistance box is connected in the right gap between C and B.
- All other connections are made as shown in the circuit diagram.
- The jockey is gently touched first to both ends of the bridge.
- Opposite deflections are observed in the galvanometer. If not, the circuit is adjusted
- The null point is located, and its position is noted.
- The current is reversed, and the null point is rechecked.
- The value of x is calculated, and readings are averaged.
- The length and diameter of the wire are measured using a meter scale and screw gauge.

EXPERIMENTAL DATA

Observations	Main scale reading (cm)	Circular scale reading (cm)	Total reading (diameter) (cm)	Mean diameter (cm)	Mean radius (cm)
1	100	37X ·001	0·037	0·037	0·0185

C. Readings for the balance point to calculate unknown resistance X.

Known resistance R (ohm)	Position of		Balance point for *l*			100-*l*	X (ohm)	Mean X (ohm)
	Unknown resistance X	Known resistance R	Direct	Reverse	Mean *l*			
6	Left	Right	29	29	29	71·	2·108	2·97
	Right	Left	29	29	29	71	2·108	
8	Left	Right	25	24·5	24·75	75·25	2·631	
	Right	Left	25	24·5	24·75	75·25	2·631	
10	Left	Right	19·2	19·1	19·15	80·85	2·370	
	Right	Left	19·2	19·1	19·15	80·85	2·370	
12	Left	Right	16·5	16·5	16·5	83·50	2.371	
	Right	Left	16·5	16·5	16·5	83·5	2·371	

Length of the wire L = 100 cm

CALCULATIONS

Calculate mean $X = \frac{X1+X2+X3+X4+X5+X6+X7+X8}{8}$

$= \frac{2.108+2.108+2.6317+2.631+2.370+2.370+2.371+2.371}{8}$

$= 2.37$

Calculation for specific resistance

Specific resistance of the material of the given wire,

$$\rho = \frac{X\pi r^2}{L}$$

$$= \frac{2.37 \times 3.1416 \times (0.185)^2}{100}$$

$= 2.548 \times 10^{-3}$ ohm-cm

Standard value of the specific resistance of the material of the given wire,

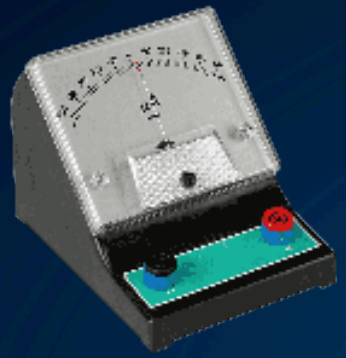
$\rho_0 = 6.09 \times 10^{-5}$ ohm.cm

Percentage error $= \frac{\rho - \rho_0}{\rho_0} \times 100\ \%$

$= \frac{2.548 \times 10^{-3} - 6.09 \times 10^{-5}}{6.09 \times 10^{-5}} \times 100$

$= 40.84\ \%$

RESULT







The value of unknown resistance X =	**2.37 Ω**
The specific resistance of the material of the given wire =	2.548 x 10^(-3)

DISCUSSION

- It should be ensured that none of the plugs in the resistance box R is loose.
- Care should be taken to determine the diameter of the wire very accurately.
- If the null point changes appreciably when the current is reversed, the thermoelectric effect is considered too large. In such a case, the galvanometer circuit should be closed while keeping the battery circuit open.

- Minor errors were introduced when the wire's diameter was measured using the screw gauge.
- Accurate location of the null point was made difficult by small galvanometer deflections.
- Resistance readings were affected by temperature fluctuations, as resistivity is temperature-dependent.
- These issues led to slight variations in the final calculation of specific resistance.

THANK YOU

